

HARSHAL ARVIND PATIL

Full Stack Developer | AI & Web Applications
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[LinkedIn](#) | [Portfolio](#) | [Google Cloud](#)

EXPERIENCE

- AI Full Stack Developer Intern (Paid) | TransformNXT | Aug 2026 - Present
- Engineered a real-time AI-powered proctoring and anti-cheat system for live video interviews and technical assessments using React, TypeScript, FastAPI, WebSockets, and MongoDB
 - Designed and deployed client-side computer vision and audio analysis pipelines that reduced server compute load while enabling 30 FPS real-time fraud detection across video, voice, and browser behavior
 - Integrated multi-modal violation scoring, automated evidence logging, and live alerting workflows into the existing assessment platform currently scaling toward production
- Full Stack Developer Intern | Vform Tecnopacks Pvt. Ltd. | ([Live Demo](#)) Jun 2026 – Jul 2026
- Built and deployed an Enterprise Asset Management System (AMS) using React, Node.js, Express, MySQL with role-based access control for Admin, Manager, and Staff
 - Delivered asset assignment, maintenance tracking, and analytics dashboards — currently in active production use by the company
- Python & AI/ML Intern (Remote) | Sure Trust | Aug 2025 – Jan 2026
- Completed a 6-month structured program covering Python, data handling, and introductory ML workflows
 - Applied concepts through hands-on assignments and a 2-month capstone project implementation
- Freelance Full Stack Developer | Self-Employed | May 2024 – Present
- Delivered 3 client projects across mobile and web — including a Java + XML Android voting app and 2 production clinic websites with WhatsApp API and Google Maps integration
 - Live: [Dental Clinic](#) | [Saikrupa Clinic](#)

PROJECTS

RecruitAI – AI-Powered Recruitment Platform ([Demo](#))

- Tech: [React](#), [FastAPI](#), [Flask](#), [Python](#), [SBERT](#), [Groq LLM](#), [MediaPipe](#), [TensorFlow.js](#), [DeepFace](#), [Whisper](#), [MySQL](#), [MongoDB](#)
- Built an end-to-end AI hiring platform reducing HR screening time by 80% using fine-tuned SBERT for semantic resume matching with SHAP-based explainable scoring
 - Developed a 3-round adaptive AI interviewer powered by Llama 3.3 with Whisper STT, truthfulness detection, and dynamic question generation
 - Engineered a real-time anti-cheat system using MediaPipe 468-point facial mesh, COCO-SSD object detection, and DeepFace identity verification

TRADEDO – AI-Powered Autonomous Stock Trading Platform

- Tech: [Python](#), [FastAPI](#), [Next.js](#), [LightGBM](#), [Groq LLM](#), [Upstox API](#), [SQLite](#), [Telegram Bot](#)
- Built a solo autonomous trading platform scanning 150+ NSE stocks across 10+ data sources with a 200-point composite scoring engine and 12-gate safety validation
 - Trained LightGBM models on 1.2M+ data points with 49 engineered features, achieving 79.65% win rate at defined confidence thresholds
 - Designed a self-improving Q-Learning system that adapts thresholds, feature weights, and position sizing based on trade outcomes

Presente – QR-Based Attendance SaaS

- Tech: [React Native](#), [Expo](#), [Node.js](#), [PostgreSQL](#), [Prisma ORM](#), [JWT](#)
- Built a production mobile app solving attendance fraud with time-expiring QR sessions per class
 - Architected a 4-role RBAC system (Super Admin, Owner, Trainer, Student) with a SaaS subscription model and 28-day billing cycles
 - Implemented automated background jobs for hourly absent-marking and generated course-wise attendance analytics

TECHNICAL SKILLS

- Programming Languages: JavaScript, Python, Java, SQL
- Frontend Development: HTML, CSS, React.js, Tailwind CSS, Basic Three.js integrations
- Backend Development: Node.js, FastAPI, Flask, Rest APIs, JWT
- Databases: PostgreSQL, MongoDB, Supabase, MySQL
- AI / ML: SBERT, LightGBM, SHAP, Whisper, spaCy, Scikit-learn, NLP
- LLMs & CV: Groq (Llama), Gemini API, MediaPipe, TensorFlow.js, DeepFace, OCR
- Cloud & Tools: Vercel, Render, Railway, Firebase, Git, GitHub, Postman

EDUCATION

B.Tech in Computer Science Engineering | Pillai College of Engineering, New Panvel 2023 - 2027
CGPA: **8.30 / 10** (till Sem VI) | Class XII (HSC): **82.17%** | Class X (CBSE): **81.50%**

ACHIEVEMENTS & ACTIVITIES

- Smart India Hackathon 2025 (SIH) — Qualified internal college round, advanced to next selection stage
- Nirman National Hackathon (Amity University Mumbai) — Built AI-powered leak management platform with GPS smart clustering as Full Stack contributor
- Participated in CODEIGNITE & Gen AI Hackathon 2024; represented college football team at Mumbai University 2024 Tournament

CERTIFICATIONS

- [Google Cloud Skills Boost](#) — Earned 22 badges across Generative AI, LLMs, Agent Development Kit (ADK), Gemini Enterprise, Transformers, NLP, and Neural Networks (Google DeepMind track)
- Full Stack Developer Bootcamp — GeeksforGeeks | Introduction to Generative AI — Google | GitHub Skills — GitHub